Designing Urban Infrastructure that Combats Air and Sound Pollution using Microalgae and Biomimicry

Introduction

According to the book "Sustainable Cities and Society", issues such as air and sound pollution are detrimental for people in urban societies due to the rapid increase in urban development (Roy 2021). Along with these health issues, the damage to the environment also builds up.

Some researchers have been inspired to look into different topics and come up with different ways to solve these issues. One of these researched topics, microalgae, has been looked into as it is capable of absorbing carbon dioxide from the atmosphere. More specifically, Chlorella vulgaris has shown great capabilities in removing carbon dioxide and other chemicals in the air with the use of photobioreactors(PBR).

Fig 1. Previous Urban Infrastructure Designs with PBR's (Cervera, 2024)

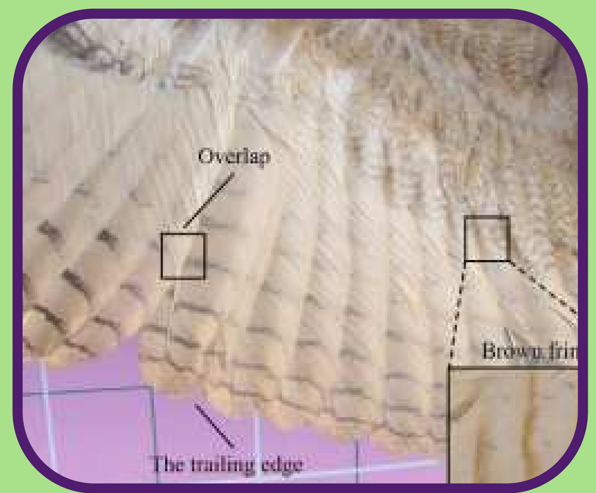

Fig 2. The wing of an eagle owl (Chen, 2012)

Additionally, some solutions for sound pollution have come up as a result of these research studies, like acoustic panels and sound proofing materials. Moreover, studies involved with sound pollution have researched into the "silent flight" of owls and attempted to imitate the similar characteristics in other settings. The carbon fiber composite of polylactic acid carbon fiber (PLA-CF) was one of the closest materials researched that could replicate the movement of the feather as it had a similar structure, density, and tensile strength (Liang et al. 2021).

Purpose

It is pertinent that these issue be solved, as many researchers think that the damage from these pollutants may end up becoming permanent soon. Solving pollution issues is vital to keeping those who live and work in cities safe from getting health issues from doing their daily activities. Using previous research about sound and air pollution, creating a piece of urban infrastructure at a smaller scale to understand its effects on its surroundings could be used for future applications in a real-life environment.

Methods

1

Fig 3. Cultivating algae

Phase 1: Algae

- Set up PBR with medium and water
- *Cultivate Chlorella vulgaris* and flocculate trials
- Find most efficient conditions and collect data

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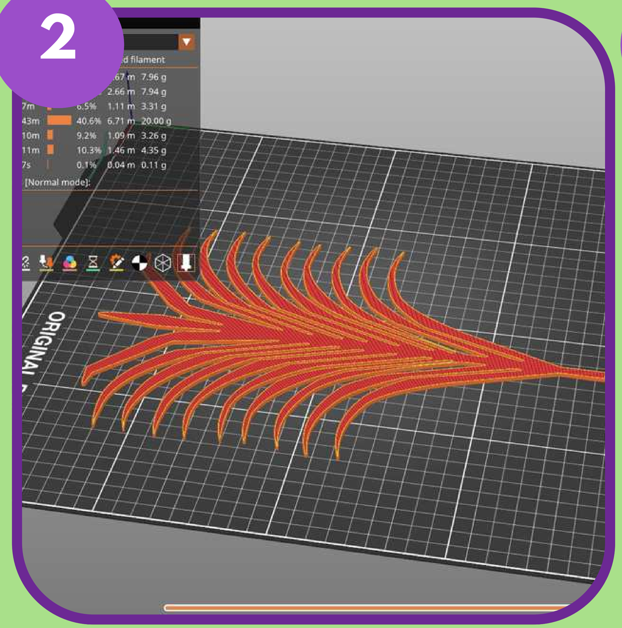

Fig 4. Photo of Inventer Owl Sketch

Phase 2: Feathers

- Sketch feather on desmos and convert points to excel
- Use points to make Inventor Sketch with Excel files
- 3D print in PLA-CF

3

Fig 5. Base Component

Phase 3: Components

- Prepare concrete and pour into base.
- Cure for a week with water spray
- Melt pellets and place into ring mold and wait for it to dry

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Fig 6. Sound Testing Apparatus

Phase 4: Testing

- Create Sound Panel Box with decibel meter and speaker
- Attach feathers to the plastic ring and attach on to the base
- Test prototype in sound setup

Results

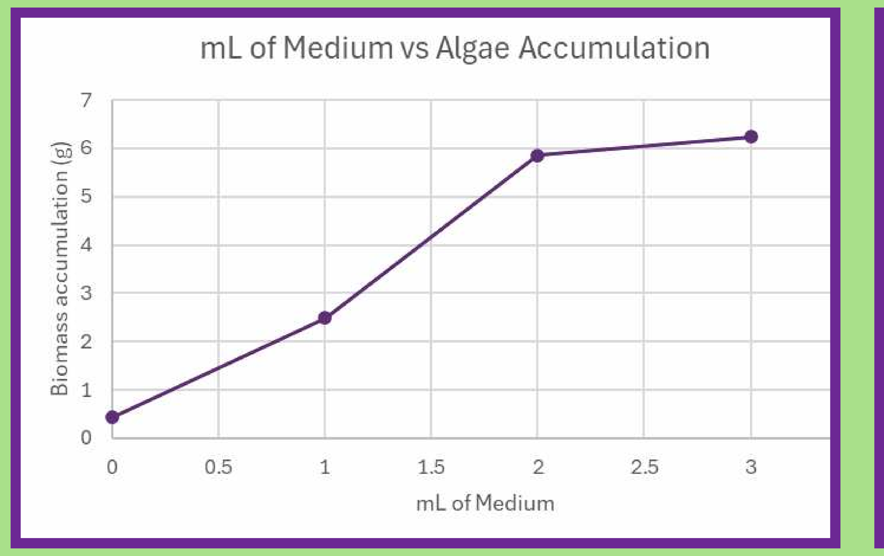

Fig 7. Algae Culture Data

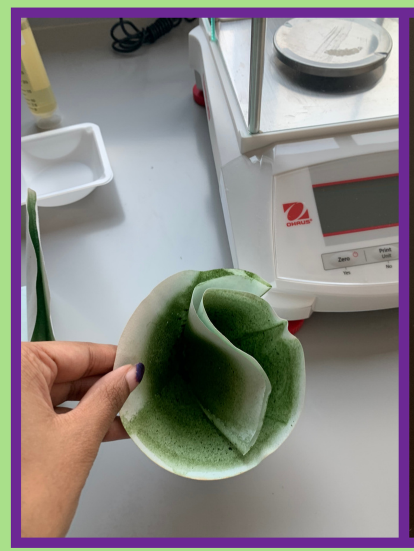

Fig 8. Algae Samples

Fig 9. Feather Prototype

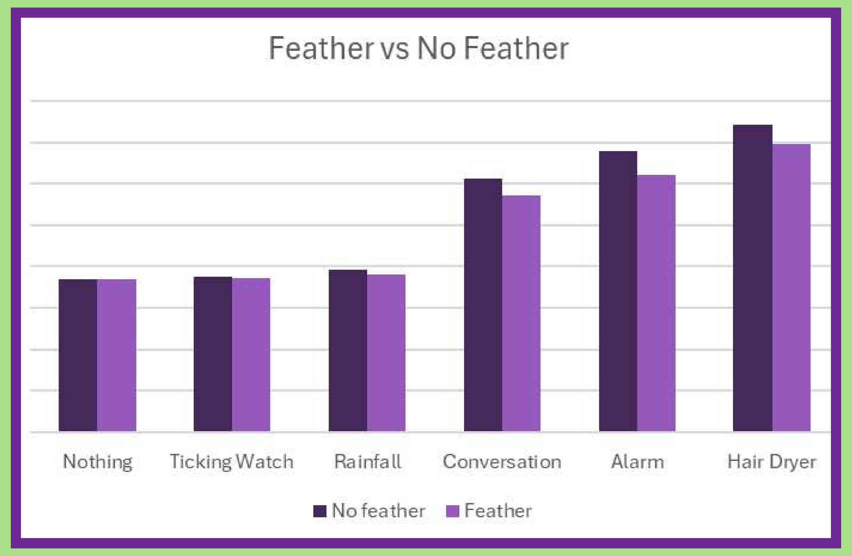

Fig 10. Decibel Meter Data

Most Efficient is at 3mL of medium in algae culturing. This trial demonstrated a growth of 6.234 grams

Least Efficient is at 0mL of medium in algae culturing. This trial demonstrated a growth of 0.4265 grams.

Fig 11. Final Prototype

Max obstruction of sound is demonstrated as 5.9 decibels for the sound similar to an alarm

Lowest obstruction of sound is demonstrated as 0.2 for the sound similar to a ticking watch.

Conclusion

The results illustrated how the 3mL of medium tested in the algae culture had the most growth, while the algae culture with no medium had the least amount of growth. This data led to 3mL of medium used in the prototype for the most efficient growth conditions

In the sound testing process, the different categories of sound consisted of items that are typically in a different range that the others. This data includes different sounds that could occur all along the urban areas, such as hairdryers and alarms, to create a realistic sound set-up. With the data collected, it seems that the feather was able to absorb sound more efficiently for louder sounds. This difference could have happened because it is harder to pick up on quieter sounds with the decibel meter. Since the decibel scale is logarithmic in growth, this reveals that a bigger amount of noise has been reduced.

Overall, the data has shown that microalgae and the biomimicry of owl structures have both aided the reduction of pollution in the environment. Finding ways to implement these components into more urban innovation could protect the environment and the people in it.

Extensions

With the success of the algae growth, creating a large scale prototype would expand on the research conducted. Additionally, trying different base designs, regarding how the microalgae is placed should be studied in order to find optimal conditions.

Implementing the use of different species pf microalgae, such as *Schedesmus sp.* and *Spirulina sp* should also be studied to understand where the strengths of each microalgae is and where they can be implemented.

Future research can look into the disposal of microalgae biomass through its source of nutrition for fertilizer, implementation in biofuels, or integration in industrial/cosmetic industries.

References

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